

SANDEEP JENA

AI ENGINEER

SUMMARY

AI Engineer applying computer vision and natural language processing to extract insights from multimodal data. Expertise in model development, deployment, and Productionization for creating impactful and scalable AI-driven solutions.

SKILLS

Programming Languages: Python, C, Bash, OpenAI Triton

RESTful APIs & RPC: Flask, FastAPI, gRPC

Image Processing: OpenCV, MediaPipe, YOLOv4/v5/v7/v8

Deep Learning & Machine Learning: Pandas, Numpy, MLflow, Scikit-learn, RAG, Nvidia Triton Inference Server
PyTorch, TensorFlow, Keras, FFmpeg, Hugging Face, Nemo, LangChain, LlamaIndex, LangGraph, LangFuse

Deployment: ONNX, TensorRT, Riva Speech SDK, OpenVINO, TensorRT-LLM, Triton, Nvidia Deepstream SDK

Development Boards: Nvidia Jetson, Raspberry Pi

Database: MongoDB, WeaviateDB, ChromaDB, InfluxDB

Cloud: AWS, Microsoft Azure Cognitive Services

DevOps: Linux, Git, Docker, Kubernetes

WORK EXPERIENCE

Inflect Technologies - AI Engineer

Bengaluru, India || 04/2024 - 04/2025

- Architected (independently designed and implemented) a robust adversarial defense system leveraging DenseNet and Resnet architecture with multi-resolution input and dynamic self-ensembling of intermediate layer predictions significantly improving model resilience against adversarial attacks and maintaining accuracy
- Developed and implemented a TensorRT-based deep learning model optimization pipeline, addressing critical CUDA out-of-memory bottlenecks, and delivering stable, high-throughput inference for various models like YOLOv5, YOLOv8, Detectron2, Patch2Pix, and custom classifiers
- Architected an autonomous monitoring system using LangGraph and Langfuse that analyzes retail data in real-time, created a RAG system with DeepSeek, Claude, Qwen and Gemini models integrated with Model Context Protocol (MCP) to process multiple data sources, designed logic to spot key business issues (like missing high-selling products), and developed smart alerts with justifications, cutting response times by 40% and improving compliance by 35%

TensorGo Technologies - Computer Vision Engineer

Hyderabad, India || 02/2023 - 10/2023

- Architecting audio/video AI platform leveraging GPT-3.5, RAG, TTS and voice cloning for animation studio
- Achieved seamless translation, transcription to 40% increased efficiency
- Architected a cutting-edge AI-powered speech analysis platform integrating offline and online speech recognition (Nemo and OpenAI), diarization (Nemo and pyannote), and advanced text processing with GPT-4 to automate meeting workflows.
- Engineered end-to-end solution for meeting summarization and insight generation, creating summaries, meeting notes, probing questions, and action items, significantly enhancing collaboration and decision-making efficiency
- Architected (designed and implemented end-to-end) AI-powered healthcare system leveraging Frozen RAG, an open-source embedding model, and a fine-tuned Llama 2 text inference engine. This system enabled efficient retrieval, summarization and generation of patient healthcare information, leading to reduced information search time and improved understanding of patient histories, thereby boosting clinician efficiency

TensorGo Technologies - Computer Vision Intern

Hyderabad, India || 08/2022 - 01/2023

- Created distracted driver detection system on Raspberry Pi achieving 7 FPS using TensorFlow and OpenCV
- Reduced infrastructure costs by 200% through optimized edge deployment
- Created (developed end-to-end) Raspberry Pi system utilizing OpenCV and TensorFlow to detect distracted driving, achieving 7 FPS throughput (sufficient for real-time edge analysis)
- Architected (designed and implemented end-to-end) deep learning model optimization pipeline using OpenVINO toolkit to run on CPUs instead of costly GPUs, reducing perinference cost by 8-10X

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in [sandeep](#)

🌐 [Github](#)

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EDUCATION

Kalinga Nagar Polytechnic

Diploma Electrical Engineering
2013-2016

Utkal University

B.SC Physics, Chemistry and Mathematics
2016-2019

COURSEWORK

iNeuron.ai

Data Science (DLCVNLPL)
sep 2021 - Aug 2022

Udemy

Tensorflow Developer Certificate in 2021:
Zero to Mastery
Oct 2021 - Aug 2022

LANGUAGE

ODIA

HINDI

ENGLISH

INTERESTS

Gaming

Chess

Self-Improvement